Flop runout cheat sheet

Exact turn and river odds for every kind of flop · Texas hold'em (and any game with a flop, turn and river)

How to read it. Turn odds count the 49 cards not on the flop; river odds ("by the river") count all 1,176 possible turn + river pairs. The board is all that's known: hole cards aren't removed, so each relevant card in your own hand moves a turn number by about 2%.

Suit and pairing odds are the same for every flop of that type, which is what makes them worth memorizing. Straight odds depend on the ranks, so they're given as averages with a range.

2.04%	9 cards	11 cards	10 cards
each turn card (1 in 49)	pair an unpaired flop: 18.4%	third flush card on two-tone: 22.4%	four-flush on monotone: 20.4%

1 · How often each flop comes

FLOP TYPE	FLOPS	HOW OFTEN	ODDS
Rainbow, unpaired	6,864	**31.1%**	1 in 3.2
Two-tone, unpaired	10,296	**46.6%**	1 in 2.1
Monotone, unpaired	1,144	**5.2%**	1 in 19.3
Rainbow, paired	1,872	**8.5%**	1 in 11.8
Two-tone, paired	1,872	**8.5%**	1 in 11.8
Rainbow, trips	52	**0.2%**	1 in 425.0
Rainbow / two-tone / monotone			39.8% / 55.1% / 5.2%
Unpaired / paired / trips			82.8% / 16.9% / 0.2%

Out of 22,100 possible flops. More than half of all flops are two-tone, and about 1 in 6 is paired. Monotone flops come about 1 time in 19; trips about 1 in 425. Every number in this sheet is exact, counted over every flop, turn and river.

2 · Flush runouts

Rainbow flop K♠ 8♥ 3♦

TURN CARD	CARDS	CHANCE	ODDS
Goes two-tone (matches a flop suit)	36	**73.5%**	1 in 1.4
Fourth suit (stays rainbow)	13	**26.5%**	1 in 3.8

BY THE RIVER	PAIRS	CHANCE	ODDS
Flush possible (runner-runner)	198	**16.8%**	1 in 5.9

Two-tone flop K♥ 8♥ 3♦

TURN CARD	CARDS	CHANCE	ODDS
Third card of the suit: flush possible	11	**22.4%**	1 in 4.5
Second card of the other suit (2nd flush draw)	12	**24.5%**	1 in 4.1
Neither suit	26	**53.1%**	1 in 1.9

BY THE RIVER	PAIRS	CHANCE	ODDS
Flush possible	539	**45.8%**	1 in 2.2
Four of the suit on board	55	**4.7%**	1 in 21.4

Monotone flop K♥ 8♥ 3♥

TURN CARD	CARDS	CHANCE	ODDS
Fourth of the suit: four-flush	10	**20.4%**	1 in 4.9
Off-suit	39	**79.6%**	1 in 1.3

BY THE RIVER	PAIRS	CHANCE	ODDS
Four or five of the suit	435	**37.0%**	1 in 2.7
Five of the suit (board flush)	45	**3.8%**	1 in 26.1

3 · Pairing runouts

Unpaired flop K 8 3

TURN CARD	CARDS	CHANCE	ODDS
Pairs the board	9	**18.4%**	1 in 5.4

BY THE RIVER	PAIRS	CHANCE	ODDS
Board paired (or better)	456	**38.8%**	1 in 2.6
Two pair on board	27	**2.3%**	1 in 43.6
Trips on board	9	**0.8%**	1 in 130.7

Paired flop K K 3

TURN CARD	CARDS	CHANCE	ODDS
Trips on board (a case king)	2	**4.1%**	1 in 24.5
Pairs the 3: two pair on board	3	**6.1%**	1 in 16.3

BY THE RIVER	PAIRS	CHANCE	ODDS
Trips (or better) on board	98	**8.3%**	1 in 12.0
Two pair on board	198	**16.8%**	1 in 5.9
Full house on board	9	**0.8%**	1 in 130.7

Trips flop K K K

TURN CARD	CARDS	CHANCE	ODDS
Quads on board	1	**2.0%**	1 in 49.0

BY THE RIVER	PAIRS	CHANCE	ODDS
Quads on board	48	**4.1%**	1 in 24.5
Full house on board	72	**6.1%**	1 in 16.3

4 · Straight runouts

Straight odds depend on how connected the ranks are, so these are averages over every unpaired flop of each kind, with the lowest and highest value in that group. “Straight possible” means three board cards fit in one five-card straight, so two hole cards can make it.

UNPAIRED FLOP	HOW OFTEN	EVENT	AVERAGE	RANGE
Straight already possible e.g. 9-8-6, J-T-7, 5-4-A	22.4%	Four to a straight on the turn	**20.9%**	16%–33%
		Four to a straight by the river	**39.1%**	30%–55%
		Five-card straight on board by the river	**2.1%**	1%–4%
Two cards connect e.g. K-8-7, Q-9-3, A-T-4	76.2%	Turn makes a straight possible	**40.7%**	24%–49%
		Straight possible by the river	**74.6%**	60%–83%
		Four to a straight by the river	**9.0%**	4%–12%
Nothing connects e.g. K-7-2, Q-7-2, A-8-3	1.4%	Straight possible by the river	**36.7%**	

Counting rule for two connecting cards

With exactly two flop cards that could share a straight, in the middle of the deck, the turn makes a straight possible with **4 cards for each rank that fits**: 24 cards when they touch, then 4 fewer for each gap.

TWO CONNECTING CARDS	TURN CARDS	TURN	BY THE RIVER
Touching (8-7) flop 2-8-7	24	**49.0%**	75.9%
One gap (9-7) flop 2-9-7	20	**40.8%**	70.9%
Two gaps (T-7) flop 2-T-7	16	**32.7%**	67.3%
Three gaps (J-7) flop 2-J-7	12	**24.5%**	59.7%

5 · One line per flop type

	TURN · FLUSH	TURN · PAIRING	RIVER · FLUSH	RIVER · PAIRING
Rainbow, unpaired 31.1% of flops	Two-tone **73.5%**	Pairs **18.4%**	Flush possible **16.8%**	Paired **38.8%**
Two-tone, unpaired 46.6% of flops	Flush possible **22.4%**	Pairs **18.4%**	Flush possible **45.8%**	Paired **38.8%**
Monotone, unpaired 5.2% of flops	Four-flush **20.4%**	Pairs **18.4%**	Four-flush+ **37.0%**	Paired **38.8%**
Rainbow, paired 8.5% of flops	Two-tone **73.5%**	Trips **4.1%**	Flush possible **16.8%**	Trips+ **8.3%**
Two-tone, paired 8.5% of flops	Flush possible **22.4%**	Trips **4.1%**	Flush possible **45.8%**	Trips+ **8.3%**
Rainbow, trips 0.2% of flops	Two-tone **73.5%**	Quads **2.0%**	Flush possible **16.8%**	Quads **4.1%**

“River” means by the river (after both the turn and river). Paired flops can also become two pair on board: 6.1% on the turn.

Memory tips

- **Count cards, then double:** on the turn each card is about 2% (exactly 2.04%). 9 cards ≈ 18%, 11 ≈ 22%, 12 ≈ 24.5%, 36 ≈ 73.5%.
- **The board pairs by the river about 39% of the time** (38.8%) on any unpaired flop, roughly 2 times in 5.
- **Two-tone flops bring a flush-possible board by the river 45.8% of the time**, nearly half; rainbow flops only 16.8% (runner-runner), about 1 in 6.
- **Monotone flops:** the turn is a fourth flush card 1 time in 5 (20.4%); by the river a fourth (or fifth) flush card lands 37% of the time.
- **Paired flops:** the case card (trips) arrives 4.1% on the turn and 8.3% by the river; the board becomes two pair 6.1% on the turn.

> **Study tip:** learn the card counts (9, 11, 10, 12, 36 on the turn) and the four “by the river” anchors: **38.8%** paired, **45.8%** two-tone flush, **16.8%** rainbow flush, **37%** monotone four-flush. Most other numbers follow from them.

Test yourself

Cover the right-hand column and answer from memory.

#	QUESTION	ANSWER (COVER ME)
1	Turn pairs an unpaired flop	**18.4% (9 cards)**
2	Unpaired flop is paired by the river	**38.8%**
3	Two-tone flop: turn is a third flush card	**22.4% (11 cards)**
4	Two-tone flop: flush possible by the river	**45.8%**
5	Rainbow flop: flush possible by the river	**16.8%**
6	Monotone flop: turn is a fourth flush card	**20.4% (10 cards)**
7	Monotone flop: four or more of the suit by the river	**37.0%**
8	Paired flop: turn makes trips on board	**4.1% (2 cards)**
9	Paired flop: trips or better by the river	**8.3%**
10	Touching cards (8-7): turn makes a straight possible	**49.0% (24 cards)**
11	Straight-possible flop: turn gives four to a straight	**20.9% on average**
12	How often is the flop two-tone?	**55.1%**
13	How often is the flop paired?	**16.9%**
14	How often is the flop monotone?	**5.2%**

Generated by tools/build-runout-pdf.js in the Poker Board Analyzer project. All figures are exact counts over 22,100 flops, 49 turn cards and 1,176 turn + river pairs per flop.